EXPERT, STEP 23

# A workflow with a stop rule

Keep fixing until the test passes. Who decides when to stop?

```
# python3 test_changes.py, before the fix
FAIL: changes.py does not print the three expected lines.
Expected:
  2. route  NL-BE    77 ->  101  +24
Got:
  2. client Epsilon Garden (example)     9 ->   20  +11
```

THE SETUP

A small script with one planted bug, a test, and two Never lines in CLAUDE.md.

THE TEST

The same ask, plain and as a workflow, 5 sessions each. Then a test that cannot pass.

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# The plain ask fixed it too

Both asks fixed the bug. The plain ask used far fewer tokens and no more questions.

```
# plain ask, the edit on line 52
-    changes.sort(key=lambda c: (-abs(c[5]), c[1], c[0]))
+    changes.sort(key=lambda c: (-abs(c[4]), c[1], c[0]))
# then the start of Claude's answer
The test passed on the second run, after one fix.
```

PLAIN ASK

Test passed in 5 of 5. 2 or 3 permission questions. About 118,000 to 162,000 tokens, most of them cache reads.

WORKFLOW ASK

Test passed in 5 of 5. 3 or 4 permission questions, plus the workflow question. About 306,000 to 361,000 tokens, most of them cache reads.

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# Read the plan before it runs

In accept edits mode, a workflow asks before it runs, and it can show you the script.

```
# the workflow script, lines 31 to 33
if (r.failing_count < best) { best = r.failing_count;
noProgress = 0 } else noProgress++
}
if (noProgress >= 2) return { status: 'stalled', rounds:
round, history }
```

THE QUESTION

Run a dynamic workflow? For a new script, option 2 shows the raw script.

THE STOP RULE

Two lines in the script: no progress counts up, and 2 rounds without progress stop the loop.

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# While it runs

The workflow runs in the background. /workflows shows each round.

```
# /workflows, after the run
fix-changes-loop
1/1 agent · 13s · done
✔ round 1       Opus 5.5 · 29.9k tok   13s
```

THE VIEW

One phase, one agent per round, with its time.

THE QUESTIONS

In accept edits mode, the workflow agent asks before most of its shell commands: twice in each one-round run here.

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# The stop rule, on a test that cannot pass

We made the test expect -25, while 48 minus 71 is -23.

```
# step 10 of the kit, from Claude's final report
The workflow has a bug: it ran 8 rounds instead of
stopping after 2 without progress.
```

PLAIN ASK

One fix, then Claude stopped by its own judgement. 2 of 2.

WORKFLOW

Of 4 runs we let finish, 3 stopped on the no-progress rule after 2 or 3 rounds. 2 of them counted the real fix as no progress. 1 ran to its cap of 8 rounds. Claude changed neither the test nor the CSV.

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# Save it, run it again

Save the script from /workflows with s. In a new session it is a slash command.

```
# after s in /workflows, then Enter
Dynamic workflow saved to
.../.claude/workflows/fix-changes-loop.js. Invoke as
/fix-changes-loop ...
```

BEFORE YOU SAY YES

- [ ] Find the loop.
- [ ] Find the stop rule.
- [ ] See how it measures progress.
- [ ] Look for a round cap.
- [ ] See which files it tells the agent to leave alone.

TRY IT

The test folder, our saved script and a step that makes the test impossible: AI Starter Kit, Expert, step 23.